

Who Says Water and Electricity Don't Mix?

by Tony Laska
and Shawn O'Neill

Oregon boasts a national reputation for being green, both in its mountain watersheds and in environmental awareness. For nearly five years, residential customers of Oregon's largest electric and gas providers have been receiving Home Energy Reviews and financial incentives to improve the efficiency of their homes. Now the process is being taken to a whole new level with the addition of water conservation services and incentives. In an innovative partnership between three Portland area water utilities and Energy Trust of Oregon, customers will reap the benefits of their collective expertise for addressing both energy and water consumption issues.

Energy Trust is an independent, nonprofit organization dedicated to energy efficiency and the development of renewable energy programs for Oregon customers of Portland General Electric, Pacific Power, NW Natural, and Cascade Natural Gas. Customers of the electric utilities pay a public purpose charge to support investments in energy efficiency and renewable energy projects, while gas utilities invest in efficiency only. According to Diane Ferington, Energy Trust's residential sector manager, the organization is always looking to leverage the efforts of other utilities to provide customers with better, more comprehensive ways to conserve resources.

Energy Trust's Home Energy Reviews, provided under the group's Home Energy Solutions program, have focused primarily on reducing consumption of electricity and gas. Now the program is evolving to include water conservation. This is a



Water use and energy consumption go hand-in-hand. This low-flow swivel kitchen aerator with shutoff reduces both.

natural extension of any conservation program, as water use and energy consumption go hand in hand—using less domestic hot water reduces the energy required to heat it.

Energy Trust's Home Energy Solutions program targets customers in homes that were built before 1985. According to Energy Trust, these homes have the greatest need for weatherization and equipment upgrades and are excellent candidates for water conservation measures to further increase savings.

Home Energy Solutions is implemented on behalf of Energy Trust by Conservation Services Group (CSG), a national energy services firm. CSG has 25 years of experience in residential conservation programs and was hired by Energy Trust in 2005. CSG conducts approximately 400 Home Energy Reviews a month for customers in Energy Trust's territory. One such customer, Sarah Murphy, was intrigued by the possibility of enhancing these reviews to encourage water conservation. Not

only that, she was in a position to do something about it, through her job as a residential water conservation coordinator with the City of Portland Water Bureau. To get the ball rolling, Murphy proposed a pilot program to her work group, secured management support and preliminary funding, and began to outline the program in detail. Water utilities work closely with each other in the Portland area, so it wasn't long before two neighboring agencies, the Rockwood Water People's Utility District and the Tualatin Valley Water District, joined the Portland Water Bureau in helping to plan the program, in cooperation with Energy Trust and CSG.

To set up this pilot program, Energy Trust is leveraging the existing infrastructure of the Home Energy Solutions program. Customers contact Energy Trust's call center to learn about the program, to determine whether they are eligible to participate, and to schedule a water assessment if they are. To determine if they qualify for a Home Energy Review and water



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Energy Advisor Kendal Hansen measures the flow rate of an existing showerhead to determine whether it qualifies for replacement with a free 1.75 gpm showerhead.

assessment, customers are asked a series of questions. How old is the home? Which water utility serves the home? Which energy utility? How many bathrooms does the home have? How did the customer hear about the program? Why is the customer interested in the program? For maximum cost-effectiveness and efficiency, a water assessment is scheduled only if it is combined with a Home Energy Review, and the review and the water assessment are conducted by the same energy advisor. To date, CSG has cross-trained eight of these advisors in both energy and water conservation strategies.

Energy advisors assigned to perform water assessments carry other materials besides those used in the standard Home Energy Review—energy-efficient CFLs, faucet aerators, and low-flow showerheads. They also carry dye tablets, and a packet of information for the customer. The packet includes information on water sources, tips on finding leaks, and tips on water conservation.

Energy advisors begin a water assessment by dropping a dye tablet into the tank of each toilet in the house to check for leaks. During a house visit, an energy advisor will also

- teach customers how to access and read their water meter and use it to determine whether there are any leaks in the house;
- record the location and severity of leaks;
- identify inefficient fixtures;
- determine whether the washing machine is Energy Star qualified and note the model of the dishwasher, if any; and
- install or leave behind water conservation devices to be installed as needed. These devices may include 1.75 gpm showerheads; 1 gpm bathroom faucet aerators; 1.5 gpm kitchen faucet aerators with a temporary shutoff switch, which interrupts flow while maintaining the setting for water temperature levels and flow rates; toilet displacement bags; toilet fill cycle diverters; and replacement toilet flappers.

At the end of the water assessment, the advisor leaves a report of recommendations with the customer. Participating water utilities are supplied with an Excel worksheet of field data for tracking and future follow-up measures.

The combination Home Energy Review and water assessment is an

experiment with a number of variables. Recognizing that only a modest number of customers could participate in this pilot, two of the water utilities began by doing their own recruiting and screening and referred candidates to Energy Trust's call center to schedule appointments. The third water utility began by marketing the pilot through its Web site. In the early stages, all utilities targeted low- to moderate-income customers. Calls from the general public increased significantly when two of the water utilities began an advertising campaign. Enrollment for the water assessment program began in late 2006, and the first assessments were performed in January 2007. As of this writing, three months into the initiative, nearly 100 water assessments have been scheduled.

CSG notes that there is huge potential for developing programs of this kind nationwide. Integrating water conservation into existing residential energy conservation programs will reduce the strain on resources and cut residents' utility bills at the same time.

Mixing water and electricity can work. Says Murphy, who came up with the idea, "This partnership is a natural fit. Working with a group that already does home energy assessments provides great value, both to our customers and to our organization. In Oregon, we are lucky to have Energy Trust setting a high standard for conservation customer service, and I think our partnership will be a great success."



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For more information:

To learn more about Energy Trust's Home Energy Solutions program, call 1-866-368-7878 or visit www.energytrust.org.

For more information about CSG and its services, visit www.conservationservicesgroup.com.